ASSIGNMENT 1

Structural Dynamics and Aeroelasticity

Images and Comments

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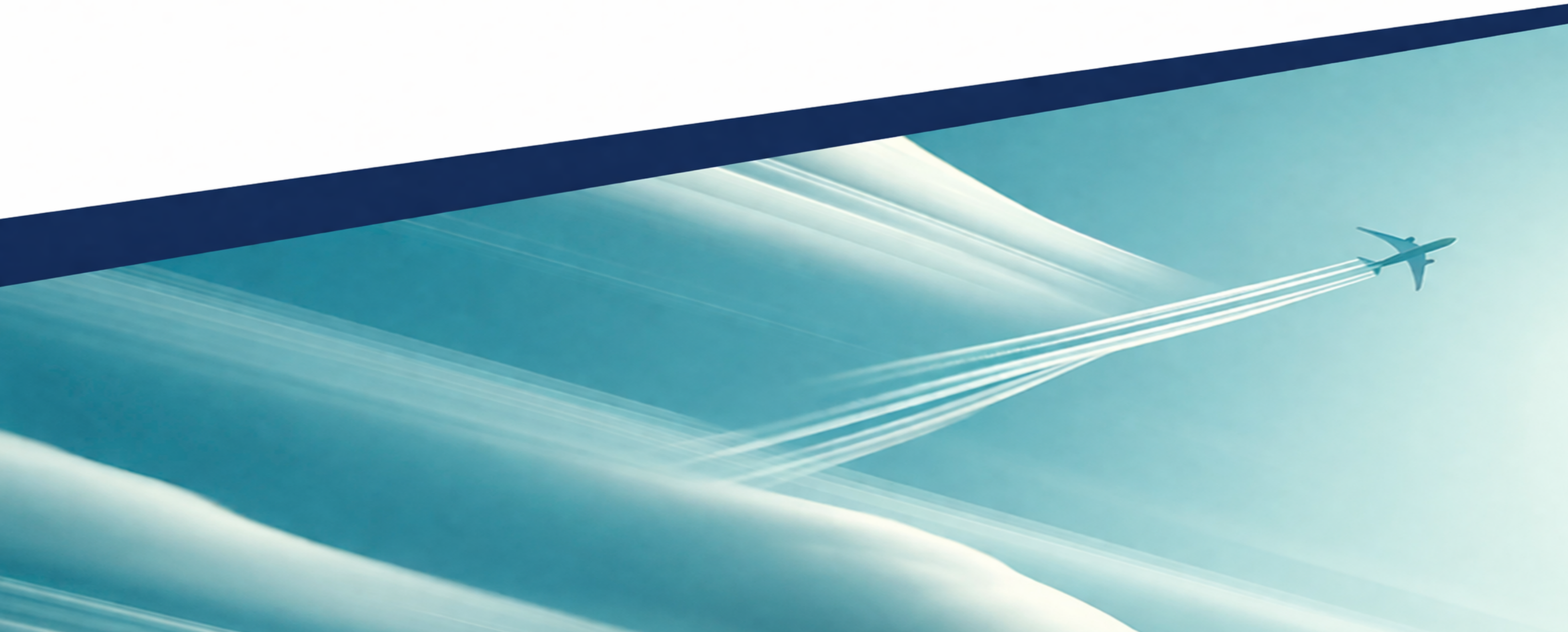


Influence of strut position on divergence

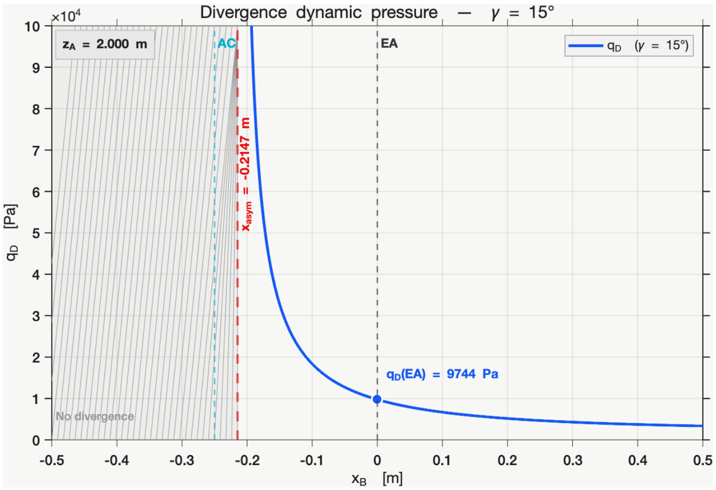


FIGURE 1 – q_D vs. x_B for $\gamma = 15°$.

q_D is highly sensitive to x_B. A vertical asymptote at $x_B \simeq -0.2147$ m separates two regimes :

- **Right :** divergence at positive q_D, dropping rapidly toward the elastic axis ($q_D \approx 9.7 \times 10^3$ Pa).
- **Left :** flexural–torsional stabilization suppresses static divergence entirely.

Limitations. No-divergence region holds within the linear model only – flutter and control reversal remain unconstrained. Forward strut placement may incur buckling/-mass penalties and is sensitive to geometric tolerances near the asymptotic boundary.

Influence of γ on the control effectiveness

x_B fixed ; $E_C = L/L_R$ with $E_C = 1$ rigid limit, $E_C > 1$ amplification, $E_C < 0$ reversal. All curves start at $E_C = 1$. Three regimes :

- **Low γ (10°, 12.5°) :** $E_C > 1$ – elastic twist amplifies aileron effect – then blocked by divergence ; displayed value is actually q_D, not q_C.
- **Intermediate (15°) :** monotonic decrease, genuine reversal at $q_C = 29\,702$ Pa.
- **High γ (17.5°, 20°) :** detrimental coupling from low q ; $q_C = 24\,685$ and $21\,427$ Pa – reversal pressure decreases with γ.

$E_C \sim [\det \mathbf{K}_{\text{TOT}}]^{-1}$; negative branch past divergence is a mathematical artefact. γ controls sign and magnitude of aeroelastic feedback : low $\gamma \to$ favorable amplification (divergence governs) ; high $\gamma \to$ early control reversal.

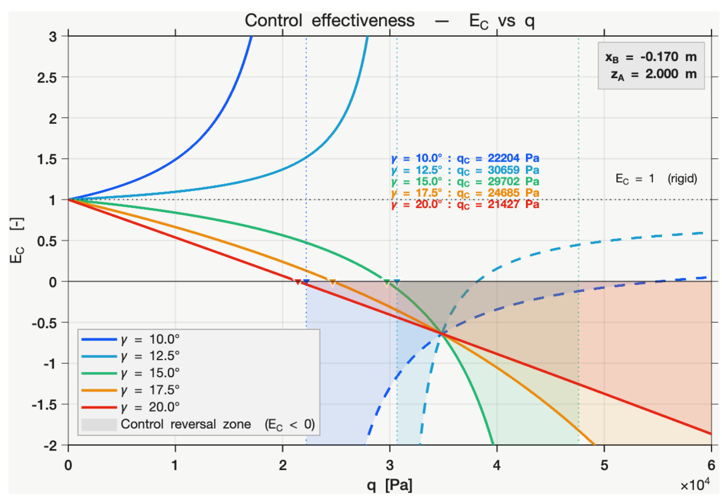
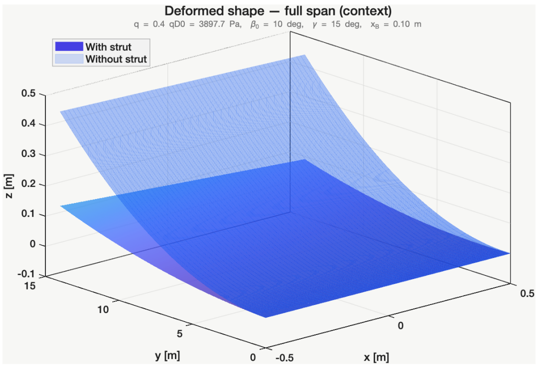


FIGURE 2 – E_C vs. q for several γ, x_B fixed.

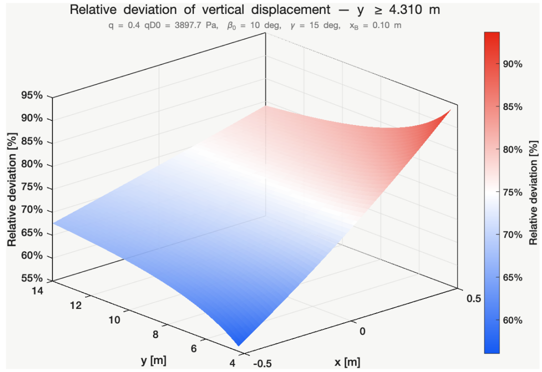
Comparison of the deformed shape with and without the strut

Config : $x_B = 0.10$ m, $\beta_0 = 10°$, $q = 0.4\,q_{D,0}$ (subcritical – differences attributable to strut stiffening, not instability proximity). δ_z interpreted only for $y \geq y_{\min} = 4.31$ m ($\approx 30.8\%\,b$), defined as the first station where the absolute gap at $x = c/4$ reaches 10% of its tip value, to avoid the root singularity ($z_{\text{ns}} \to 0$).

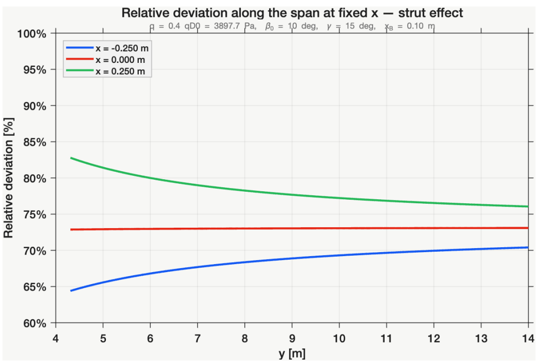
Both configurations share the same bending–torsion pattern (zero at root, increasing toward tip) ; the strut reduces amplitude without altering shape :



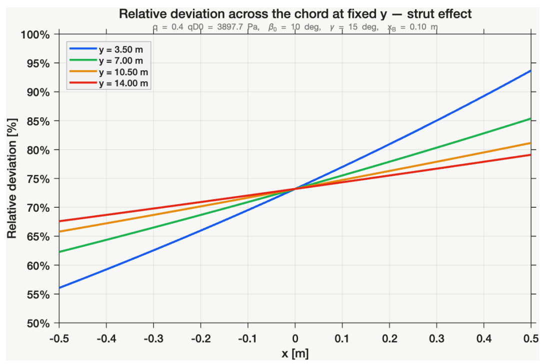
3D shape, with vs. without strut.



δ_z map, $y \geq y_{\min}$.



Spanwise cuts of δ_z.



Chordwise cuts of δ_z.

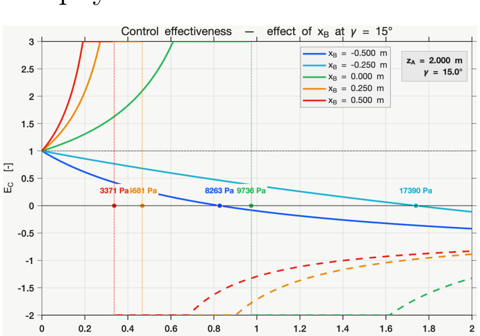
1) $\delta_z \in\sim [55\%, 95\%]$ over valid span – strut wing is $\frac{1}{4}$–$\frac{1}{3}$ as deformed as no-strut reference.

2) Along elastic axis $(x = 0) : z = (y/b)^2\phi_w$, $\delta_z \approx 73\%$ – dominant bending reduction.

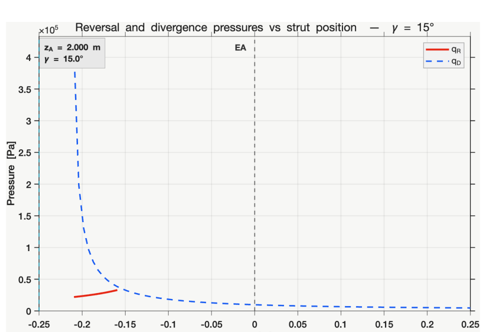
3) Chordwise variation of δ_z consistent with torsional term $-x(y/b)\phi_\theta$ – strut simultaneously stiffens bending and torsion.

Effect of the strut on the reversal dynamic pressure

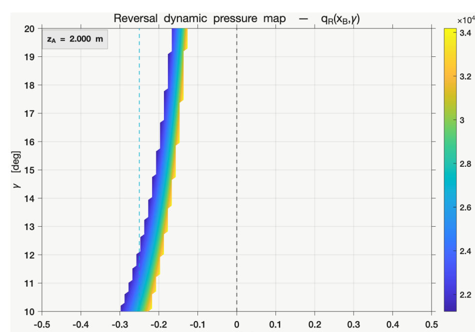
Physical reversal : first genuine zero of E_C with $0 < q_R < q_D$; sign changes near divergence singularity are not physical reversals.



E_C vs. q, several x_B.



q_R, q_D vs. x_B.

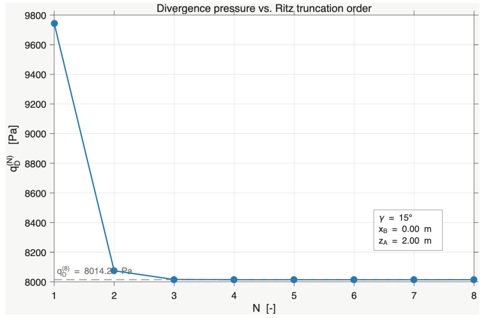


Map $q_R(x_B, \gamma)$.

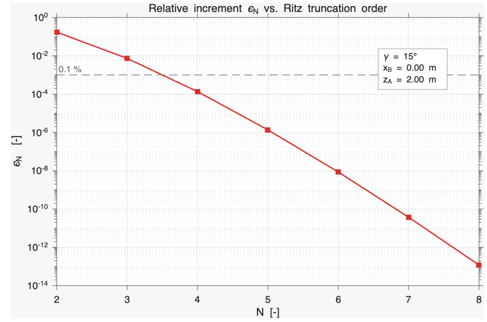
$x_B \geq 0$ **:** E_C stays positive up to divergence – no physical reversal ; apparent sign changes near asymptotes are not genuine q_R. $x_B < 0$ **:** strut modifies bending–torsion coupling ; genuine q_R may appear. **Combined** q_R/q_D **:** beneficial configuration must raise both thresholds ; physical q_R exists only over a limited band of negative x_B. **Map :** q_R confined near EA/AC for $x_B < 0$; increasing γ raises q_R – stronger coupling delays reversal. **Design objective :** maximize q_R or suppress reversal within flight envelope while keeping q_D high ; strut beneficial only for forward attachment with sufficiently large γ.

Convergence analysis of the Ritz approximation

$q_D^{(N)}$ drops sharply at low N then converges at $q_D \approx 8.014 \times 10^3$ Pa ; no visible change beyond $N = 3$. ε_N falls by many orders of magnitude, reaching $\sim 10^{-14}$ at $N = 8$ – numerical saturation at machine precision, not additional physical accuracy.



$q_D^{(N)}$ *vs.* N.



ε_N *vs.* N.

$N = 1$ overpredicts q_D by $\approx 22\%$: useful for trends but not quantitatively reliable. $N = 3$–4 already excellent ; $N = 8$ merely confirms asymptotic convergence.